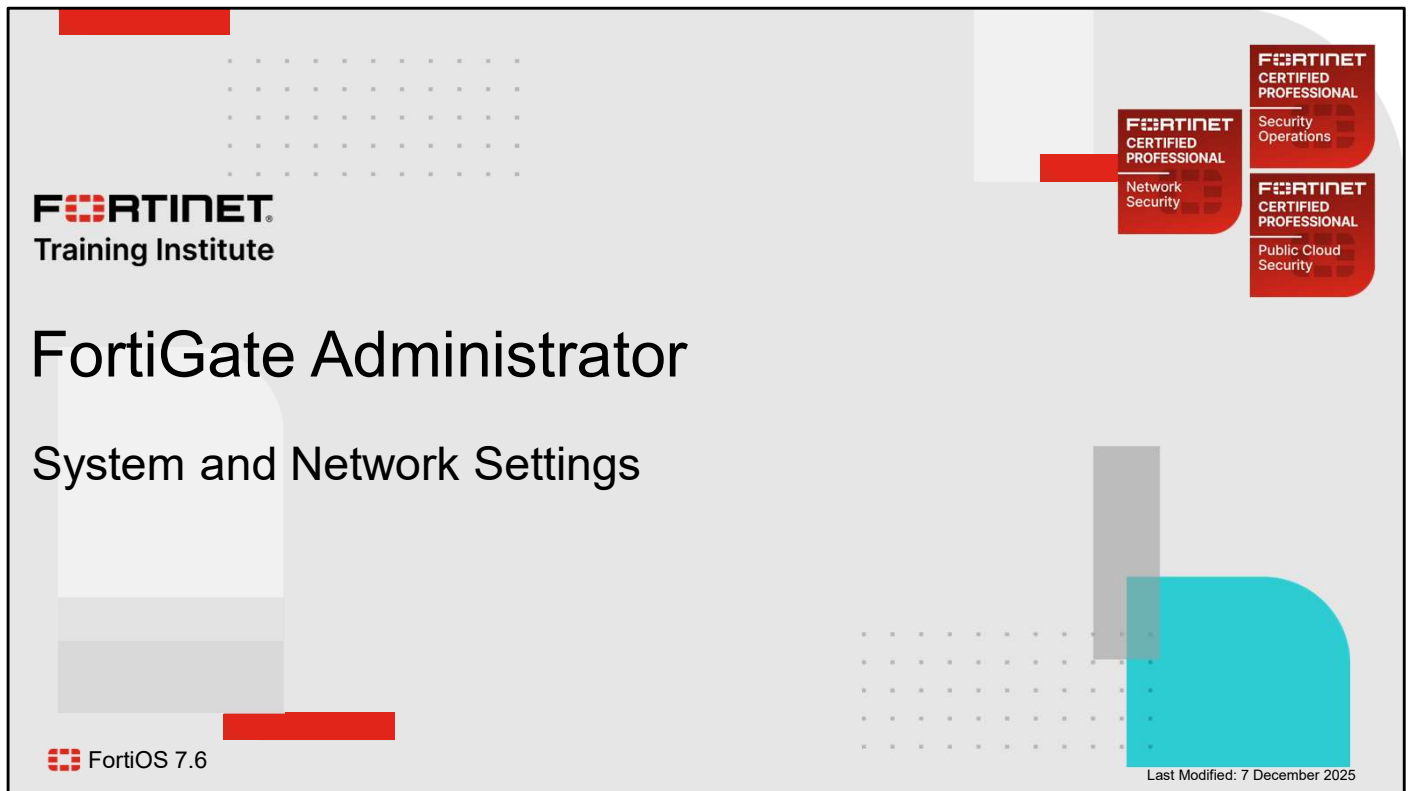
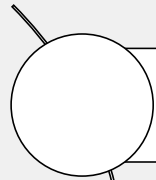


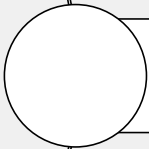


In this lesson, you will learn about system and network settings on FortiGate.

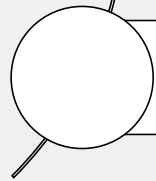
Lesson Overview



Initial Setup



Basic Administration



Fundamental Maintenance

In this lesson, you will learn about the topics shown on this slide.

Initial Setup

Objectives

- Configure FortiGate using the factory default settings
- Configure FortiGate as the DHCP server

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in basic system and network administration, you will be able to perform the initial setup of FortiGate and configure basic networking settings.

The Modern Context of Network Security

- Firewalls are more than gatekeepers on the network perimeter
- Today's firewalls are designed in response to multifaceted and multidevice environments with no identifiable perimeter:
 - Mobile workforce
 - Partners accessing your network services
 - Public and private clouds
 - Internet of Things (IoT)
 - Bring your own device (BYOD)
- Firewalls are expected to perform different functions within a network
 - Different deployment modes:
 - Distributed enterprise firewall
 - Next-generation firewall (NGFW)
 - Internal segmentation firewall (ISFW)
 - Data-center firewall
 - DNS, DHCP, web filter, intrusion prevention system (IPS), and so on

In the past, the most common method of protecting a network was securing the perimeter and installing a firewall at the entry point. Network administrators trusted everything and everyone inside the perimeter.

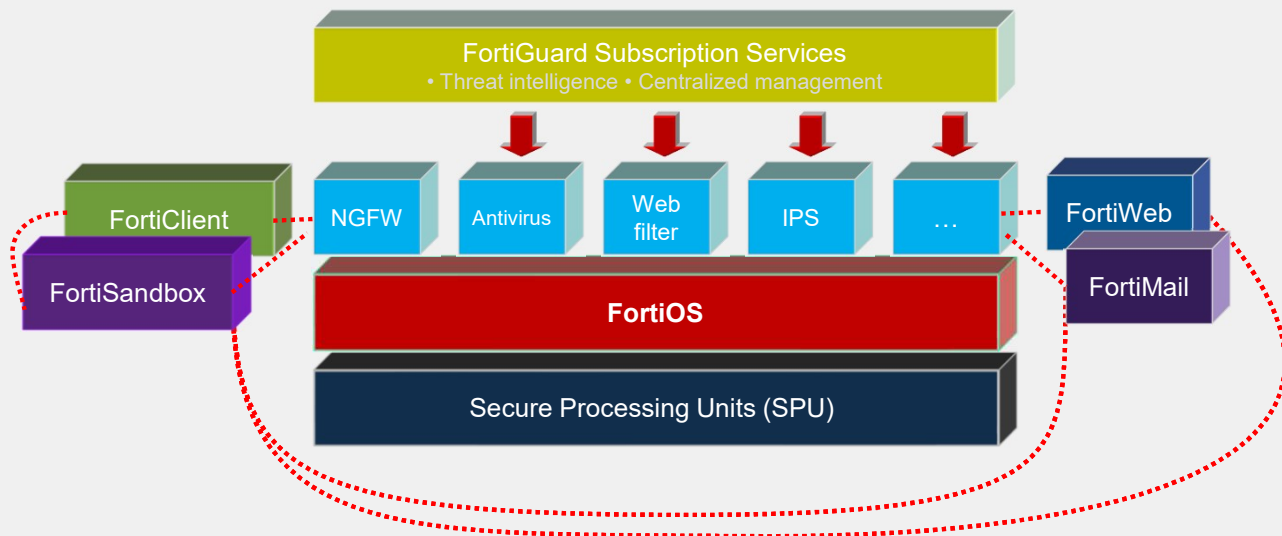
Now, malware can easily bypass any entry-point firewall and get inside the network. This could happen through an infected USB stick or a compromised device being connected to the corporate network. Additionally, because attacks can come from inside a network, network administrators can no longer inherently trust internal all users and devices.

What's more, today's networks are highly complex environments whose borders are constantly changing. Networks run vertically from the LAN to the internet, and horizontally from the physical network to the private virtual network and to the cloud. A mobile and diverse workforce (employees, partners, and customers) accessing network resources, public and private clouds, the IoT, and BYOD programs all conspire to increase the number of attack vectors faced by your network.

In response to this highly complex environment, firewalls have become robust, multifunctional devices that counter an array of threats to your network. FortiGate can act in different modes or roles to address different requirements. For example, FortiGate can be deployed as a data-center firewall whose function is to monitor inbound requests to servers and to protect them without increasing latency for the requester. Or, FortiGate can be deployed as an ISFW as a way to contain a network breach.

FortiGate can also function as DNS and DHCP servers and be configured to provide web filter, antivirus, and IPS services.

Platform Design



In the architecture diagram shown on this slide, you can see how FortiGate platforms add strength, without compromising flexibility. Like separate, dedicated security devices, FortiGate is still *internally* modular. Plus:

- Devices add duplication. Sometimes, dedication *doesn't* mean efficiency. If it's overloaded, can one device borrow free RAM from nine others? Do you want to configure policies, logging, and routing on 10 separate devices? Does 10 times the duplication bring you 10 times the benefit, or is it a hassle? For small and midsize businesses (SMB) or enterprise branch offices, unified threat management (UTM) is often a superior solution compared to separate, dedicated appliances.
- FortiGate hardware isn't just off-the-shelf. It's carrier grade. Most FortiGate models have one or more specialized circuits, called application-specific integrated circuits (ASIC), that are engineered by Fortinet. For example, a content processor (CP) or network processor (NP) chip handles cryptography and packet forwarding more efficiently. Compared to a single-purpose device with only a CPU, FortiGate can have dramatically better performance. This is especially critical for data centers and carriers where throughput is business critical.
(The exception? Virtualization platforms—VMware, Citrix Xen, Microsoft, or Oracle Virtual Box—have general-purpose virtual CPUs (vCPUs). But, virtualization might be worthwhile because of other benefits, such as distributed computing and cloud-based security.)
- FortiGate is flexible. If all you need is fast firewalling and antivirus, FortiGate will not cause you to waste CPU, RAM, and electricity on other features. In each firewall policy, you can enable or disable UTM and NGFW modules. Also, you won't pay more to add VPN seat licenses later.
- FortiGate cooperates. A preference for open standards instead of proprietary protocols means less vendor lock-in and more choice for system integrators. And, as your network grows, FortiGate can leverage other Fortinet products, such as FortiSandbox and FortiWeb, to distribute processing for deeper security and optimal performance—a total Security Fabric approach.

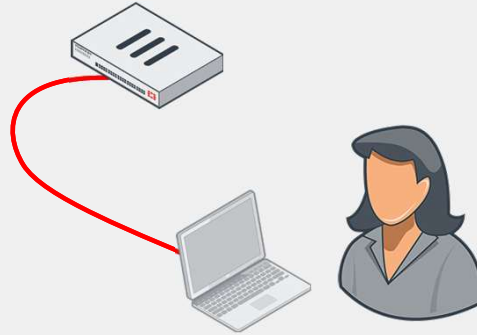
Factory Default Settings

- IP: 192.168.1.99/24
 - Management interface on high-end and mid-range models
 - Port1 or internal interface on entry-level models
- PING, HTTPS, and SSH protocol management enabled
- Built-in DHCP server is enabled on port1 or internal interface
 - Only on entry-level models that support DHCP server
- Default login

User: admin

Password: (blank)

 - Both are case sensitive
 - Modify the default (blank) password
- Can access FortiGate on the CLI
 - Console: without network
 - CLI console widget and terminal emulator, such as PuTTY or Tera Term



Network address translation (NAT) mode is the default operation mode. What are the other factory default settings? After you have removed FortiGate from its box, what do you do next?

Attach your computer network cable to port1 or the internal switch ports (on the entry-level model). For high-end and mid-range models, connect to the management interface. In most entry-level models, there is a DHCP server on that interface. So, if your computer's network settings have DHCP enabled, your computer should automatically get an IP address, and you can begin to set up.

To access the GUI on FortiGate or FortiWiFi, open a web browser and visit `https://192.168.1.99`.

The default login information is public knowledge. Never leave the default password blank. Your network is only as secure as your FortiGate admin account. Once you log in with default login details, you'll see a message instructing you to the default admin user password. Before you connect FortiGate to your network, you should set a complex password. You will also be asked to apply additional configuration information such as host name, dashboard setup, FortiCare registration, and so on.

All FortiGate models have a console port, or USB management port, or both. The port provides CLI access, without a network. You can access the CLI using the CLI console widget on the GUI or from a terminal emulator, such as PuTTY or Tera Term.

Interface IPs

- In NAT mode, you can't use interfaces until they have an IP address:
 - Manually assigned
 - Automatic
 - DHCP
 - PPPoE

Network > Interfaces

Edit Interface

Name **port5**

Alias

Type ☒ Physical Interface

VRF ID

Role

☐ Dedicated Management Port

Address

Addressing mode **Manual** IPAM DHCP PPPoE One-Arm Sniffer

IP/Netmask

Secondary IP address ☐

Network > Interfaces

Edit Interface

Name **port5**

Alias

Type ☒ Physical Interface

VRF ID

Role

☐ Dedicated Management Port

Address

Addressing mode **DHCP** IPAM PPPoE One-Arm Sniffer

Retrieve default gateway from server ☒

Distance

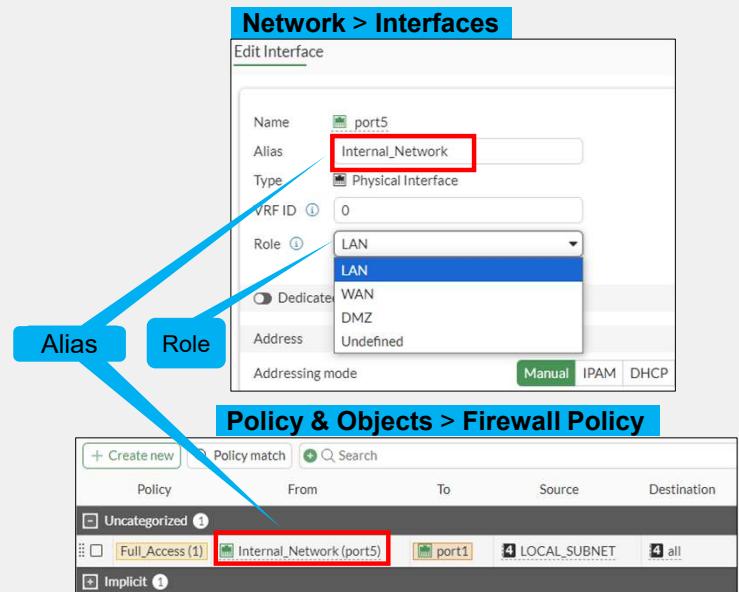
Override internal DNS ☒

When FortiGate is operating in NAT mode, every interface that handles traffic must have an IP address. When in NAT mode, FortiGate can use the IP address to source the traffic, if it needs to start or reply to a session, and as a destination address for devices trying to contact FortiGate or route traffic through it. There are multiple ways to get an IP address:

- Manually
- Automatically, using either DHCP or Point-to-Point Protocol over Ethernet (PPPoE) (available on the CLI)

Interface Role Compared to Alias

- Role defines interface settings typically grouped together:
 - Prevents accidental misconfiguration
 - Four types:
 - LAN
 - WAN
 - DMZ
 - Undefined (show all settings)
 - Not in a list of policies
- Alias is a friendly descriptor for the interface:
 - Used in a list of policies to label interfaces by purpose



How many times have you seen network issues caused by a DHCP server—not client—enabled on the WAN interface?

You can configure the interface role. The roles shown on the GUI are the usual interface settings for that part of a topology. Settings that do not apply to the current role are hidden on the GUI. (All settings are always available on the CLI, regardless of the role.) This prevents accidental misconfiguration.

For example, when the role is configured as **WAN**, there is no DHCP server and device detection configuration available. Device detection is usually used to detect devices internally on your LAN.

If there is an unusual case, and you need to use an option that's hidden by the current role, you can always switch the role to **Undefined**. This displays all options.

To help you remember the purpose of each interface, you can give them aliases. For example, you could call port3 `internal_network`. This can help to make your list of policies easier to comprehend.

FortiGate as a DHCP Server

Network > Interfaces

Edit Interface

Name: port3
 Alias: LAN
 Type: Physical Interface
 VRF ID: 0
 Role: LAN
 Addressing mode: **Manual** | IPAM | DHCP
 IP/Netmask: 10.0.1.254/255.255.255.0
 Create address object matching subnet: ☒
 Name: port3 address
 Destination: 10.0.1.0/24
 Secondary IP address: ☐

Administrative Access

IPv4: ☒ HTTPS ☒ HTTP ☒ PING
☐ FMG-Access ☐ SSH ☐ SNMP
☐ FTM ☐ RADIUS Accounting ☐ Security Fabric Connection
☐ Speed Test
 Receive LLDP: Use VDOM Setting **Enable** Disable
 Transmit LLDP: Use VDOM Setting **Enable** Disable

DHCP Server

DHCP status: **Enabled** Disabled
 Address range: 10.0.1.1-10.0.1.253
 Netmask: 255.255.255.0
 Default gateway: Same as Interface IP Specify
 DNS server: Same as System DNS Same as Interface IP Specify
 Lease time: 604800 second(s)
☒ Advanced

Wireless clients are not the only ones that can use FortiGate as their DHCP server.

For an interface (such as port3), select **Manual**, enter a static IP address, and then enable **DHCP Server**. Options for the built-in DHCP server appear, including provisioning features, such as DHCP options and IP address assignment rules.

Advanced DHCP options allow you to automatically configure clients with additional network settings, such as DNS servers or NTP configurations. You can also enable DHCP relay to forward requests to an external server which is useful when managing IP addresses across multiple subnets. Additionally, custom IP address assignment rules provide precise control by mapping specific IP addresses to MAC addresses, or allocating address ranges, preventing conflicts, and ensuring consistency across the network.

VLANs



- *Logically* subdivide your physical layer 2 network into smaller segments
 - Each segment forms a separate broadcast domain
 - VLAN tags added to frames identify their network segments

VLANs split your physical LAN into multiple, logical LANs. In NAT operation mode, each VLAN forms a separate broadcast domain. Multiple VLANs can coexist on the same physical interface, provided they have different VLAN IDs. In this way, a physical interface is split into two or more logical interfaces. A tag is added to each Ethernet frame to identify the VLAN to which it belongs. In transparent mode, VLANs operate like switches, with no IP translation. This is suitable for scenarios where invisible filtering or bridging between VLANs is needed.

Creating VLANs

- Frames sent or received by the physical interface segment are never tagged; they belong to the *native* VLAN

Network > Interfaces

FortiGate VM64-KVM

+ Create new

Interface

Zone

Virtual Wire Pair

Internal_Network (port5) Physical Interface

LAN (port3) Physical Interface

New Interface

Name: VLAN101

Alias: Branch-Firmware

Type: VLAN

VLAN protocol: 802.1Q 802.1AD

Interface: port1

VLAN ID: 101

VRF ID: 0

Virtual domain: root

Role: LAN

Address

Addressing mode: Manual IPAM DHCP PPoE One-Arm Sniffer

IP/Netmask: 0.0.0.0/0.0.0.0

Create address object matching subnet: On

Name: VLAN101 address

Destination: 0.0.0.0/0

Secondary IP address: Off

To create a VLAN using the GUI, click **Create New**, select **Interface**, and then, in the **Type** field, select **VLAN**. You must specify the VLAN ID and the physical interface to which the VLAN will be bound. Frames that belong to interfaces of that type are always tagged. On the other hand, frames sent or received by the physical interface segment are never tagged. They belong to what is called the *native* VLAN (VLAN ID 0).

You can use 802.1Q when the network requires basic VLAN segmentation with up to 4094 VLANs. 802.1ad (QinQ) is ideal when you need create to greater VLAN scalability by stacking VLANs.

Note that in a multi-VDOM environment, the physical interface and its VLAN subinterface can be in separate VDOMs.

Static Gateway

- Set a default route or configure a static one
- If the interface is DHCP or PPPoE, the gateway can be set dynamically

Network > Static Routes

Local-FortiGate

Dashboard

Network

Interfaces

DNS

IPAM

SD-WAN

Static Routes

Policy Routes

+ Create new

Destination

0.0.0.0/0

New Static Route

Destination [?] Subnet Internet Service

0.0.0.0/0.0.0.0

Gateway Address 0.0.0.0

Interface port1

Administrative Distance [?] 10

Comments Write a comment...

Status Enabled Disabled

Advanced Options

Priority [?] 1

Before you integrate FortiGate into your network, you should configure a default gateway.

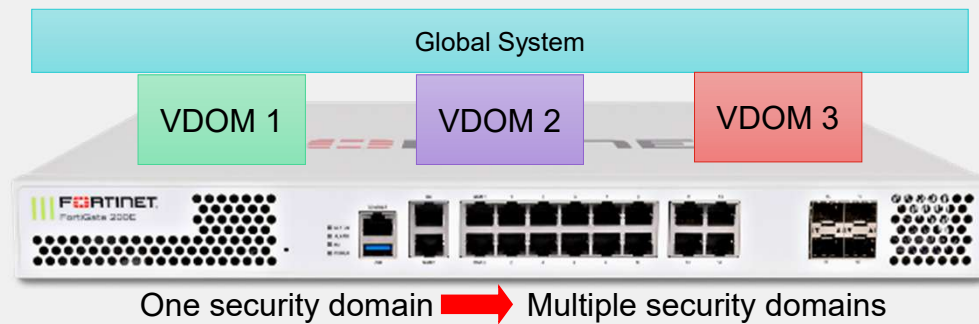
If FortiGate gets its IP address through a dynamic method, such as DHCP or PPPoE, then it should also retrieve the default gateway.

Otherwise, you must configure a static route. Without this, FortiGate will not be able to respond to packets outside the subnets directly attached to its own interfaces. It likely will not be able to connect to FortiGuard—which is crucial for FortiGate updates—and may fail to route traffic correctly.

Make sure that FortiGate has a default route (matching all packets with the destination 0.0.0.0/0) configured to forward traffic through the network interface connected to the internet, directing it to the IP address of the next router.

You must configure routing to establish essential network settings before you can configure firewall policies.

VDOMs



- VDOMs split FortiGate into multiple virtual devices
 - They employ independent security policies, routing tables, and so on
- Packets are confined to same VDOM
- Almost all FortiGate models support up to 10 VDOMs by default
 - High-end and virtual models allow for the purchase of additional VDOMs

What if, in addition to segmenting your network, you want to subdivide policies and administrators into multiple security domains?

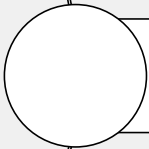
If so, you can enable FortiGate VDOMs in order to split FortiGate into multiple logical devices. Each VDOM has independent security policies and routing tables. By default, traffic from one VDOM cannot go to a different VDOM. This means that two interfaces in different VDOMs can share the same IP address, without any overlapping subnets. However, it is possible to configure routes to pass traffic between VDOMs, if that is required.

When you use VDOMs, a single FortiGate becomes a virtual data center of network security, unified threat management (UTM) inspection, and secure communication devices.

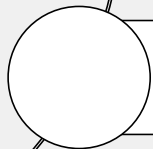
Lesson Progress



Initial Setup



Basic Administration



Fundamental Maintenance

Good job! You now understand how to perform the initial setup of FortiGate.

Now, you will learn about basic administration.

Basic Administration

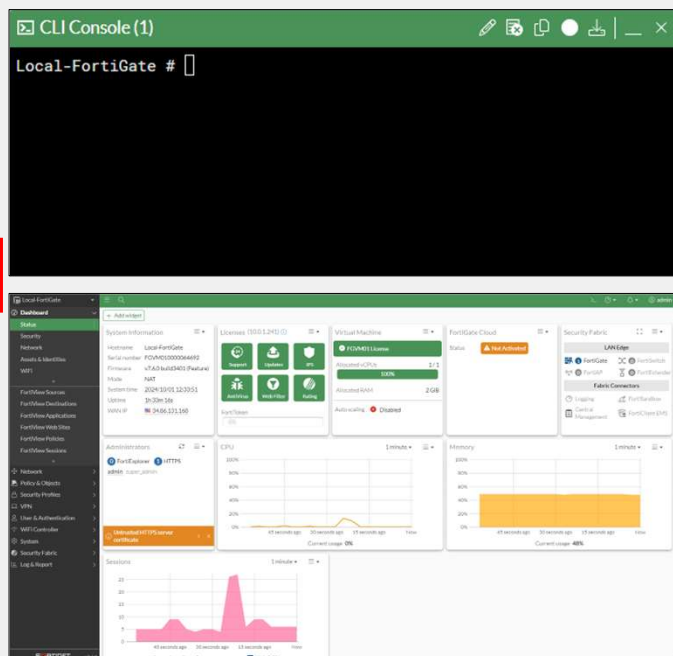
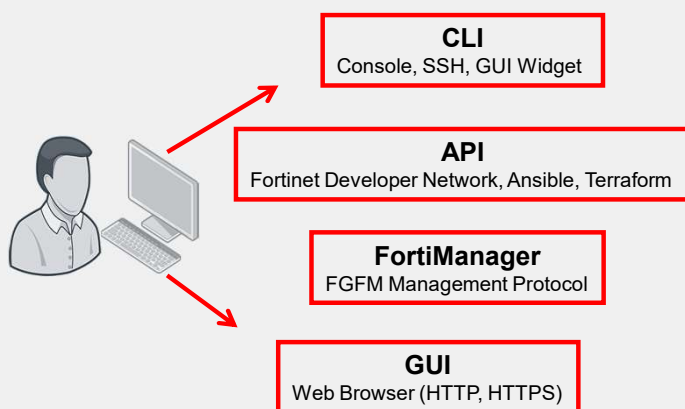
Objectives

- Configure and control administrator access to FortiGate

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in basic administration, you will be able to better manage administrative users and implement stronger security practices for administrative access.

Administration Methods



Most features are available on both the GUI and CLI, but there are a few exceptions. You can't view reports on the CLI. Also, advanced settings and diagnostic commands for super users are usually not available on the GUI.

As you become more familiar with FortiGate, and especially if you want to script its configuration, you might want to use the CLI, API or FortiManager in addition to the GUI. You can access the CLI through either the JavaScript widget on the GUI named **CLI Console**, or through a terminal emulator such as Tera Term or PuTTY. Your terminal emulator can connect through the network—SSH—or the local console port.

SNMP and some other administrative protocols are also supported, but they are read-only. You can't use them for basic setup.

Create an Administrative User

The screenshot shows the FortiGate web interface. On the left, the 'System > Administrators' menu is visible. The 'Create new' dropdown menu is highlighted with a red box, and 'Administrator' is selected. On the right, the 'New Administrator' form is shown. The 'Username' field is set to 'Administrator'. The 'Type' dropdown is set to 'Local User'. The 'Password' and 'Confirm Password' fields are highlighted with a red box. The 'Comments' field is set to 'Write a comment...'. The 'Administrator profile' dropdown is set to 'super_admin'. The 'Two-factor Authentication', 'Restrict login to trusted hosts', and 'Restrict admin to guest account provisioning only' options are all disabled.

Whichever method you use, start by logging in as admin. Begin by creating separate accounts for other administrators. For security and tracking purposes, it is a best practice for each administrator to have their own account.

In the **Create New** field, you can select either **Administrator** or **REST API Admin**. Typically, you will select **Administrator** and then assign an **administrator profile**, which specifies that user's administrative permissions. You could select **REST API Admin** to add an administrative user who would use a custom application to access FortiGate with a REST API. The application would allow you to log in to FortiGate and perform any task that your assigned **administrator profile** permits.

Other options, not shown here, include:

- Instead of creating accounts on FortiGate, you could configure FortiGate to query a remote authentication server.
- In place of passwords, your administrators could authenticate using digital certificates that are issued by your internal certification authority server.

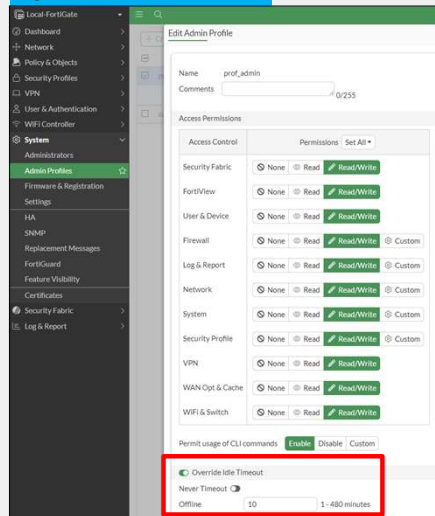
If you do use passwords, ensure that they are strong and complex. For example, you could use multiple interleaved words with varying capitalization, and randomly insert numbers and punctuation. Do not use short passwords, or passwords that contain names, dates, or words that exist in any dictionary. These are susceptible to brute force attack. To audit the strength of your passwords, use tools such as L0phtcrack (<http://www.l0phtcrack.com/>) or John the Ripper (<http://www.openwall.com/john/>). Risk of a brute force attack is increased if you connect the management port to the internet.

In order to restrict access to specific features, you can assign permissions.

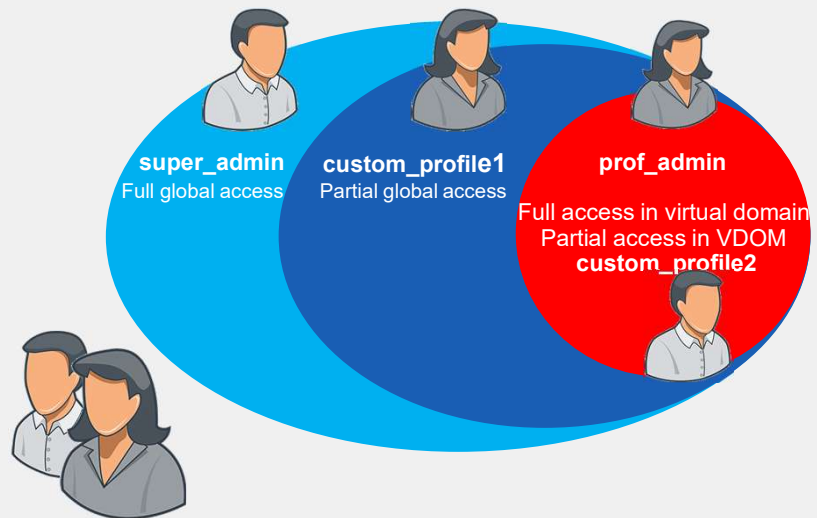
Administrator Profiles

- Permissions

System > Admin Profiles



- Hierarchy



When assigning permissions to an administrator profile, you can specify read-and-write, read-only, or none to each area.

By default, there is a special profile named `super_admin`, which is used by the account named `admin`. You can't change it. It provides full access to everything, making the `admin` account similar to a root superuser account. The `prof_admin` profile is another default profile that provides full access. However, unlike the `super_admin` profile, it provides access only to the firewall settings of the virtual domain(s) and not to the global settings of FortiGate. Also, you can modify the permissions of the `prof_admin` profile.

You aren't required to use a default profile. You could create a profile named `auditor_access` with read-only permissions. Restricting a person's permissions to those necessary for his or her job is a best practice, because even if that account is compromised, the compromise to your FortiGate device (or network) is not total. To do this, create administrator profiles and then select the appropriate profile when configuring an account.

The **Override Idle Timeout** option allows the `admintimeout` value, under `config system accprofile`, to be overridden for each access profile. You can configure administrator profiles to increase inactivity timeout and facilitate use of the GUI for central monitoring. Note that you can do this on a per-profile basis, to prevent the option from being unintentionally set globally. So, what are the effects of administrator profiles?

It's actually more than just read or write access. Depending on the type of administrator profile that you assign, an administrator may not be able to access the entire FortiGate device. For example, you could configure an account that can view only log messages. Administrators may not be able to access global settings outside their assigned virtual domain either. VDOMs are a way of subdividing the resources and configurations on a single FortiGate. Administrators with a smaller scope of permissions cannot create, or even view, accounts with more permissions.

Administrative Access—Trusted Sources

System > Administrators

☐ Two-factor Authentication

☒ Restrict login to trusted hosts

Trusted Host 1

☐ Restrict admin to guest account provisioning only

+ Create new

+ Search

☐	Na...	Trusted Hosts	Profile	Type	Two-factor Authentication
System Administrator 1					
☐	admin	10.0.1.10/32	super_admin	Local	Disabled

You can restrict an admin user to manage guest users with a guest group in place to provision users

If the admin user attempts to log in to the FortiGate GUI from any IP address other than 10.0.1.10, they receive this message

! Authentication failure

Another way to secure FortiGate is to define the hosts or subnets that are trusted sources from which to log in.

In this example, 10.0.1.10 is configured as the only trusted IP address for sources that the administrator logs in from. If the administrator attempts to log in from a machine with any other IP address, they will receive an authentication failure message.

Note that if trusted hosts are configured on all administrators and an administrator is trying to log in from an IP address that is not set on any of the trusted hosts for any administrators, then the administrator will not get the login page.

If you leave any IPv4 address as 0.0.0.0/0, it means that connections from any source IP address will be allowed. By default, 0.0.0.0/0 is the configuration for the administrator, although you may want to change this.

Note that each account can define its management host or subnet differently. Be aware of any NAT that occurs between the specified device and FortiGate. You can easily prevent an administrator from logging in from the specified IP address if it is later NATed to another address before reaching FortiGate, thus defeating the purpose of the trusted hosts.

You can also configure an administrator account that is restricted to provisioning guest accounts only. This option requires you to configure a guest user group.

Administrative Access—Ports and Password

- Port numbers are customizable
- Using only secure access (SSH, HTTPS) is recommended
- Default **Idle timeout** is 5 minutes

System > Settings

The screenshot shows the 'System > Settings' page. The 'Administration Settings' section includes fields for HTTP port (80), Redirect to HTTPS (disabled), HTTPS port (443), HTTPS server certificate (self-sign), SSH port (22), and Telnet port (23). The 'Idle timeout' is set to 5 minutes. The 'Password Policy' section includes a Password scope dropdown (set to Admin), Minimum length (8), Minimum number of new characters (0), Character requirements (disabled), Allow password reuse (enabled), and Password expiration (disabled).

You may also want to customize the administrative protocols port numbers.

You can choose whether to allow concurrent sessions. You can disallow concurrent sessions to avoid accidentally overwriting settings. Allowing concurrent settings lets an administrator open multiple browser tabs and CLI sessions at the same time. This may be more convenient for some situations, but it may also result in an administrator accidentally editing the same settings in conflicting ways over different connections.

For better security, use only secure protocols, and enforce password complexity and changes.

The **Idle timeout** settings specify the number of minutes before an inactive administrator session times out (the default is 5 minutes). A shorter idle timeout is more secure, but increasing the timer can help reduce the chance of administrators being logged out while testing changes.

You can override the idle timeout setting for each administrator profile using the **Override Idle Timeout** setting.

You can configure an administrator profile to increase inactivity timeout and facilitate use of the GUI for central monitoring. The **Override Idle Timeout** setting allows the **admintimeout** value, under **config system accprofile**, to be overridden for each access profile.

Note that you can do this for each profile individually, to avoid the option from being unintentionally set globally.

Administrative Access—Protocols

- Enable acceptable management protocols on each interface independently:
 - Separate IPv4 and IPv6
 - IPv6 options hidden by default
- Other protocols for which FortiGate is the destination IP address:
 - Security Fabric connection:
 - CAPWAP
 - FortiTelemetry
 - FMG-Access
 - FTM
 - RADIUS Accounting
- LLDP support
 - Detecting an upstream Security Fabric FortiGate through LLDP

Network > Interfaces

Edit Interface

Name  port4

Alias

Type  Physical Interface

VRF ID  0

Role  Undefined

Address

Addressing mode Manual IPAM DHCP PPPoE

IP/Netmask

Secondary IP Address 

Administrative Access

IPv4

☐ SCIM	☒ HTTPS	☒ HTTP
☒ PING	☐ FMG-Access	☒ SSH
☐ SNMP	☐ FTM	☐ RADIUS Accounting
☐ Security Fabric Connection 	☐ Speed Test	

Receive LLDP  Use VDOM Setting Enable Disable

Transmit LLDP  Use VDOM Setting Enable Disable

You've defined the management subnet—that is, the trusted hosts—for each administrator account. How do you enable or disable management protocols?

This is specific to each interface. For example, if your administrators connect to FortiGate only from port3, then you should disable administrative access on all other ports. This prevents brute force attempts and also insecure access. Your management protocols are HTTPS, HTTP, PING, and SSH. By default, the HTTP option is not visible on the GUI.

Consider the location of the interface on your network. Enabling PING on an internal interface is useful for troubleshooting. However, if it's an external interface (in other words, exposed to the internet), then the PING protocol could expose FortiGate to a denial of service (DoS) attack. You should disable protocols that do not encrypt data flow, such as HTTP. IPv4 and IPv6 protocols are separate. It's possible to have both IPv4 and IPv6 addresses on an interface, but only respond to pings on IPv6.

Security Fabric connection includes CAPWAP and FortiTelemetry. Protocols like FortiTelemetry are *not* for administrative access, but, like GUI and CLI access, they are protocols where the packets have FortiGate as a destination IP. Use the FortiTelemetry protocol specifically for managing FortiClient and the Security Fabric. Use the CAPWAP protocol for FortiAP, FortiSwitch, and FortiExtender when they are managed by FortiGate. Use the FMG-Access protocol specifically for communicating with FortiManager when that server is managing multiple FortiGate devices. Use the RADIUS accounting protocol when FortiGate needs to listen for and process RADIUS accounting packets for single sign-on (SSO) authentication. FTM, or FortiToken Mobile push, supports second-factor authentication requests from a FortiToken mobile app. When you assign the interface roles LAN or WAN to the appropriate interfaces, your FortiGate uses the Link Layer Discovery Protocol (LLDP) to detect if there's an upstream FortiGate in your network. If FortiGate discovers an upstream FortiGate, you're prompted to configure the upstream FortiGate device to join the Security Fabric.

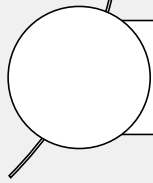
Lesson Progress



Initial Setup



Basic Administration



Fundamental Maintenance

Good job! You now understand basic administration.

Now, you will learn about fundamental maintenance.

Fundamental Maintenance

Objectives

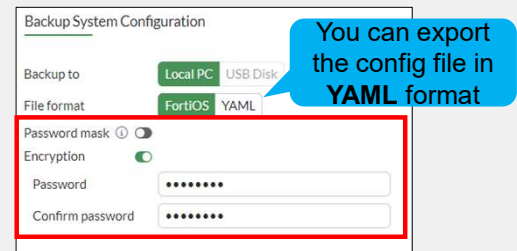
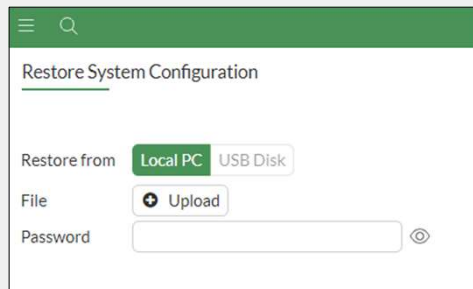
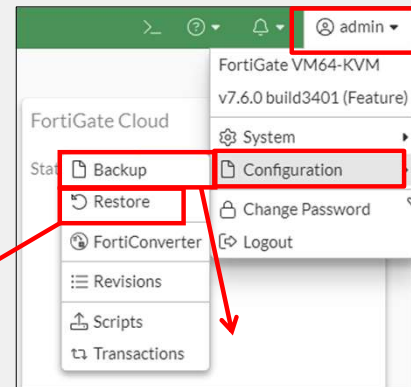
- Back up and restore system configuration files
- Upgrade FortiGate firmware
- Check and verify FortiGuard licenses

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in basic maintenance, you will be able to perform the vital activities, ensuring that FortiGate remains reliably in service throughout its life cycle.

Configuration File—Backup and Restore

- Configuration can be saved to an external device
 - It can mask passwords and secrets
 - Optional encryption
 - Can back up automatically
 - Upon logout
 - Not available on all models
- To restore a previous configuration, upload file
 - Reboots FortiGate



Now that FortiGate has basic network settings and administrative accounts, you will learn how to back up the configuration. In addition to selecting the destination of the backup file, you can choose to encrypt or not to encrypt the backup file. Even if you choose not to encrypt the file, which is the default, the passwords stored in the file are hashed, and, therefore, obfuscated. The passwords that are stored in the configuration file would include passwords for the administrative users and local users, and preshared keys for your IPsec VPNs. It may also include passwords for the Fortinet Single Sign-On (FSSO) and LDAP servers.

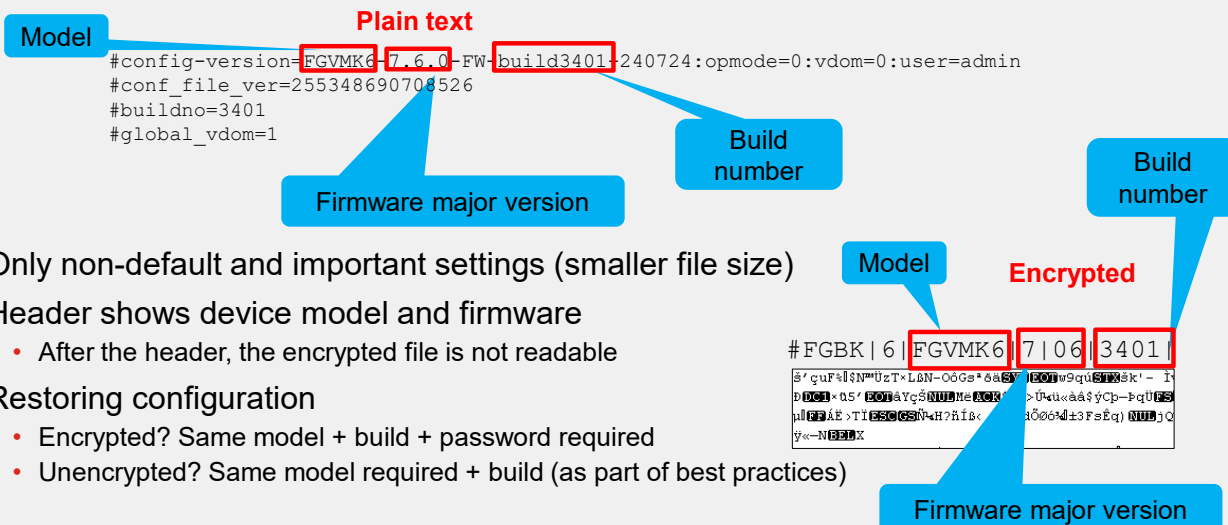
The other option is to encrypt the configuration file with a password. Besides securing the privacy of your configuration, it also has some effects you may not expect. After encryption, the configuration file cannot be decrypted without the password and a FortiGate of the same model and firmware. This means that if you send an encrypted configuration file to Fortinet technical support, even if you give them the password, they cannot load your configuration until they get access to the same model of FortiGate. This can cause unnecessary delays when resolving your ticket. Instead, you can enable password masking when creating a new backup file to replace all passwords and secrets in the configuration file and prevent unintentional data leaks when sharing the backup file with a third party.

If you enable VDOMs, subdividing the resources and configuration of your FortiGate device, each VDOM administrator can back up and restore their own configurations. You don't have to back up the entire FortiGate configuration, however, it is still recommended.

Backups help speed up the return to production if FortiGate is damaged in an unforeseen disaster.. Having to recreate hundreds of policies and objects from scratch takes a significant amount of time, while loading a configuration file on a new device takes much less.

The process for restoring a configuration file is very similar to that of creating one. The restore process will cause FortiGate to restart. Note that restoring the configuration of a single VDOM in multi-VDOM mode will only restart the processes of that VDOM—other VDOMs will be unaffected.

Configuration File Format



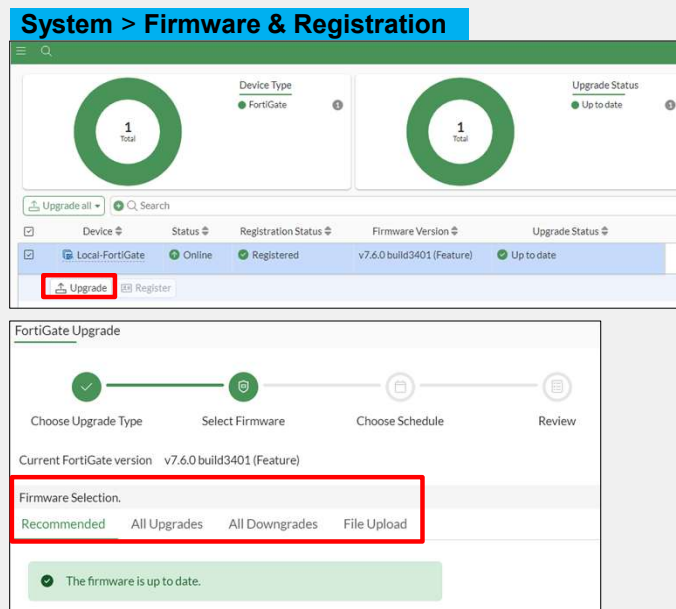
If you open the configuration file in a text editor, you'll see that both encrypted and unencrypted configuration files contain a cleartext header that contains some basic information about the device. The example on this slide shows which information is included. To restore an encrypted configuration, you must upload it to a FortiGate device of the same model and firmware, then provide the password.

To restore an unencrypted configuration file, you are required to match only the FortiGate model. If the firmware is different, FortiGate will attempt to upgrade the configuration. This is similar to how it uses upgrade scripts on the existing configuration when upgrading firmware. However, it is still recommended to match the firmware and build on FortiGate to the firmware listed in the configuration file, as best practice.

Usually, the configuration file contains only non-default settings, plus few default, yet crucial, settings. This minimizes the size of the backup, which could otherwise be several megabytes in size.

Upgrade Firmware

- You can view the current firmware version on the dashboard or in **System > Firmware & Registration** (or on the CLI: `get system status`)
- If there is an updated firmware version, you are notified
- You can update the firmware by clicking **Upgrade** and then selecting the **All Upgrades** or **File Upload** option
- Make sure you read the *FortiOS Release Notes* to verify the upgrade path and other details



You can view the current firmware version in multiple places on the FortiGate GUI. When you first log in to FortiGate, the landing page is the dashboard. You can see the firmware version in the **System** widget. This information is also found at **System > Firmware & Registration**. You can also retrieve the information on the CLI using the command `get system status`.

If a new version of the firmware is available, you are notified on the dashboard and on the **Firmware & Registration** page. The **Firmware & Registration** page allows administrators to manage the firmware running on each FortiGate, FortiAP, and FortiSwitch in the Security Fabric, and to authorize and register these Fabric devices.

You can use the **Upgrade** option to upgrade firmware of the selected device. The **Fabric Upgrade** option upgrades firmware for the root FortiGate as well as Fabric devices. You can also use this option to upgrade firmware for a non-Security Fabric FortiGate with managed FortiSwitch and FortiAP devices. The **Fabric Upgrade** option uses released firmware images from FortiGuard.

You can also use the **Register** option to register a selected device to FortiCare and an **Authorize** option to authorize a selected device for use in security fabric.

Remember to read the *FortiOS Release Notes* to make sure that you understand the supported upgrade path. The *FortiOS Release Notes* also provide pertinent information that may affect the upgrade.

FortiGuard Subscription Services

- Internet connection and contract required
- Provided by FortiGuard Distribution Network (FDN)
 - Major data centers in North America, Asia, and Europe
 - Or, from FDN through your FortiManager
 - FortiGate prefers the data center in nearest time zone, but will adjust by server load
- Package updates: FortiGuard antivirus and IPS
 - `update.fortiguards.net`
 - TCP port 443 (SSL)
- Live queries: FortiGuard web filtering, DNS filtering, and antispy
 - `service.fortiguards.net` for proprietary protocol on UDP port 53 or 8888
 - `securewf.fortiguards.net` for HTTPS over port 443, 53 or, 8888
- FortiOS uses FortiGuard server for DNS request
 - By default, uses DNS over TLS (DoT) to secure DNS traffic



Some FortiGate services connect to other servers, such as FortiGuard, in order to work. FortiGuard Subscription Services provide FortiGate with up-to-date threat intelligence. FortiGate uses FortiGuard by:

- Periodically requesting packages that contain a new engine and signatures
- Querying the FDN on an individual URL or host name

By default, the FortiGuard server location is set to anywhere FortiGate selects a server based on server load, from any part of the world. However, you have the option to change the FortiGuard server location to USA. In this case, FortiGate selects a USA-based FortiGuard server.

Queries are real-time; that is, FortiGate asks the FDN every time it scans for spam or filtered websites. FortiGate queries, instead of downloading the database, because of the size and frequency of changes that occur to the database. Also, you can select queries to use UDP or HTTPs for transport; the protocols are not designed for fault tolerance, but for speed. So, queries require that your FortiGate device has a reliable internet connection.

Packages, like antivirus and intrusion prevention system (IPS), are smaller and don't change as frequently, so they are downloaded (in many cases), only once a day. They are downloaded using TCP for reliable transport. After the database is downloaded, their associated FortiGate features continue to function, even if FortiGate does not have reliable internet connectivity. However, you should still try to avoid interruptions during downloads—if your FortiGate device must try repeatedly to download updates, it can't detect new threats during that time.

When using FortiGuard servers for DNS, FortiOS uses DNS over TLS (DoT) by default to secure the DNS traffic. New FortiGuard DNS servers have been added as primary and secondary servers.

FortiGuard Subscription Services (Contd)

- FortiGuard third party SSL certificate verification and OCSP stapling check
 - FortiGuard servers query the CA OCSP responder every four hours
- Anycast as default FortiGuard access mode
 - Optimize the routing performance to the FortiGuard servers
 - FortiGate gets a single IP address for the domain name of each FortiGuard service
 - Enforce a connection to use protocol HTTPS and port 443

Now, third-party SSL certificate verification and OCSP stapling check has been implemented for all FortiGuard servers. By default, the FortiGuard access mode is *anycast* on FortiGate, to optimize the routing performance to the FortiGuard servers. The FortiGuard server has one IP address to match its domain name. FortiGate connects with a single server address, regardless of where the FortiGate device is located.

The domain name of each FortiGuard service is the common name in the certificate of that service. The certificate is signed by a third-party intermediate certificate authority (CA). The FortiGuard server uses the Online Certificate Status Protocol (OCSP) stapling technique, so that FortiGate can always validate the FortiGuard server certificate efficiently. FortiGate will complete the TLS handshake only with a FortiGuard server that provides a *good* OCSP status for its certificate. Any other status results in a failed SSL connection.

The FortiGuard servers query the OCSP responder of the CA every four hours and update its OCSP status. If FortiGuard is unable to reach the OCSP responder, it keeps the last known OCSP status for seven days.

FortiGate aborts the connection to the FortiGuard server if:

- The common name (CN) in the server certificate does not match the domain name resolved from the DNS.
- The OCSP status is not good.
- The issuer-CA is revoked by the root-CA.

The FortiGuard access mode *anycast* setting forces the routing process to use protocol HTTPS, and port 443.

FortiGuard Subscription Services (Contd)

- Some of the FortiGuard domain names and their IP addresses:

Server	Domain name and IP address
Object download	globalupdate.fortinet.net - 173.243.140.6
Querying service (webfiltering, antispy)	globalguardservice.fortinet.net - 173.243.140.16
FortiGate Cloud logging	globallogctrl.fortinet.net - 173.243.132.25
FortiGate Cloud management	globalmgrctrl.fortinet.net - 173.243.132.26
FortiGate Cloud messaging	globalmsgctrl.fortinet.net - 173.243.132.27
FortiGate Cloud sandbox	globalaptctrl.fortinet.net - 184.94.112.22
The productapi used by OCVPN registration and GUI icon download	globalproductapi.fortinet.net - 66.35.17.252

The table on this slide shows a list of some of the FortiGuard servers and their domain names and IP addresses.

FortiGuard Licenses

System > FortiGuard

FortiGuard Distribution Network

License Information

Entitlement	Status	
Advanced Malware Protection	Licensed (Expiration Date: 2026/01/17)	
Attack Surface Security Rating	Licensed (Expiration Date: 2026/01/17)	
Data Loss Prevention (DLP)	Not Licensed	
Email Filtering	Licensed (Expiration Date: 2026/01/17)	
Intrusion Prevention	Licensed (Expiration Date: 2026/01/17)	
Operational Technology (OT) Security Service	Not Licensed	Purchase
Web Filtering	Licensed (Expiration Date: 2026/01/17)	
SD-WAN Network Monitor	Not Licensed	Purchase
SD-WAN Overlay as a Service	Not Licensed	Purchase
FortiSASE SPA Service Connection	Not Licensed	Purchase
FortiSASE Secure Edge Management	Not Licensed	Purchase
FortiGate Cloud	Not Activated	Activate
FortiGate Cloud Sandbox	Licensed (Expiration Date: 2026/01/17)	
FortiToken Cloud	In Trial (Expiration Date: 2024/10/08)	Upgrade
Firmware & General Updates	Licensed (Expiration Date: 2026/01/17)	
FortiCare Support	Registered	Actions
FortiConverter Service	Not Licensed	
Virtual Machine	Valid	FortiGate VM License

FortiCare support contracts can be activated here and applied directly to this FortiGate.

Enter Registration Code

FortiGuard Updates

Scheduled updates

Every / Daily / Weekly / Automatically

FortiGuard Updates

Next Update: 2024/10/01 13:31:00

Update Licenses & Definitions Now

Manual Update

Upload License File

Fortinet Service Communications

Service	Traffic Volume (Last 24 hours)
FortiCare	0 B
FortiGate Cloud Log	0 B
FortiGuard.com	556.11 kB
FortiGuard Download	8.89 kB
FortiGuard Query	7.77 kB
FortiGate Cloud Sandbox	0 B
SDNS	0 B
FortiToken Registration	0 B
SMS Service	0 B

Additional Information

API Preview

Edit in CLI

Online Guides

Relevant Documentation

Video Tutorials

How to Purchase/Renew Fortinet Service Subscriptions

You can check the status of FortiGuard licenses and the communication to FortiGuard on the FortiGate GUI. You can also check the versions of the locally installed databases for each of the FortiGuard services.

FortiGuard Licenses (Contd)

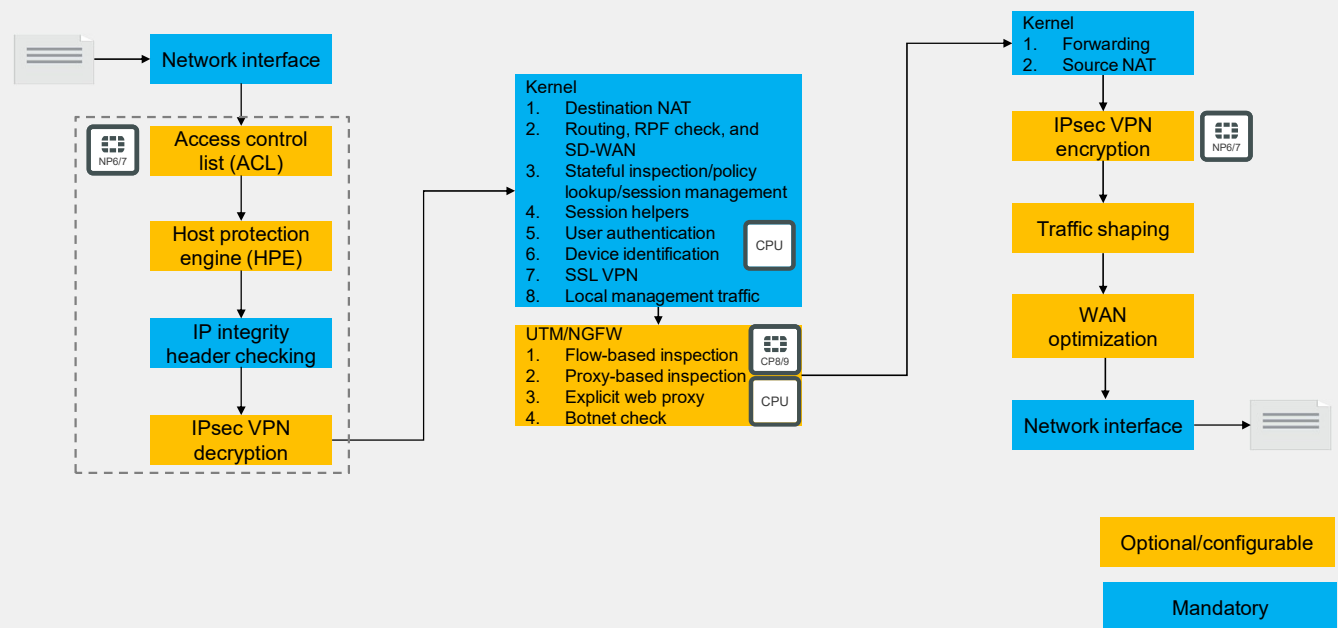
```
Local-FortiGate # diagnose autoupdate versions
AV Engine
-----
Version: 7.00030 signed
Contract Expiry Date: Mon Jan 19 2026
Last Updated using manual update on Thu Jul 13 02:54:00 2024
Last Update Attempt: Mon Aug 25 13:52:18 2024
Result: No Updates

Virus Definitions
-----
Version: 90.01635 signed
Contract Expiry Date: Mon Jan 19 2026
Last Updated using manual update on Mon Jul 25 13:52:18 2024
Last Update Attempt: Mon Aug 25 13:52:18 2024
Result: Updates Installed
```

The command shown on this slide lists all the FortiGuard databases and engines installed. The information includes the version, contract expiration date, time it was updated, and what happened during the last update.

The list includes but is not limited to antivirus, IPS, application, mobile malware definitions, and other security services. FortiGate is licensed and updated using FortiGuard services.

Life of a Packet—Initial Session Packets



This slide shows the steps that the first packets of a session go through as they enter, pass through, and exit FortiGate.

FortiGate first performs some security inspections such as access control list (ACL), Host Protection Engine (HPE), and IP integrity header checking, to make sure the packets are within acceptable parameters before allowing them to move through the rest of the processes. These inspections are handled by the network processor (NP) to minimize impact on the FortiGate CPU.

Each version of the NP has criteria that define which traffic can be offloaded. The NP enhances overall performance by allowing offloaded sessions to bypass the FortiGate CPU after the session is established and the session key is installed on the NP. The NP can also handle IPsec VPN encryption and decryption operations, where the configured encryption and hashing algorithms are supported in the hardware.

The content processor (CP) functions like a coprocessor for the FortiGate CPU to improve overall system performance by offloading certain tasks, such as pattern matching for flow-based UTM inspection with the IPS engine, SSL/TLS decryption and encryption for deep SSL inspection, and IPsec encryption and decryption operations for supported algorithms.

Note that the packet processing for virtual FortiGate devices is almost identical with the only difference being that the CPU handles all processes instead of being able to offload some of them to NPs and CPs.

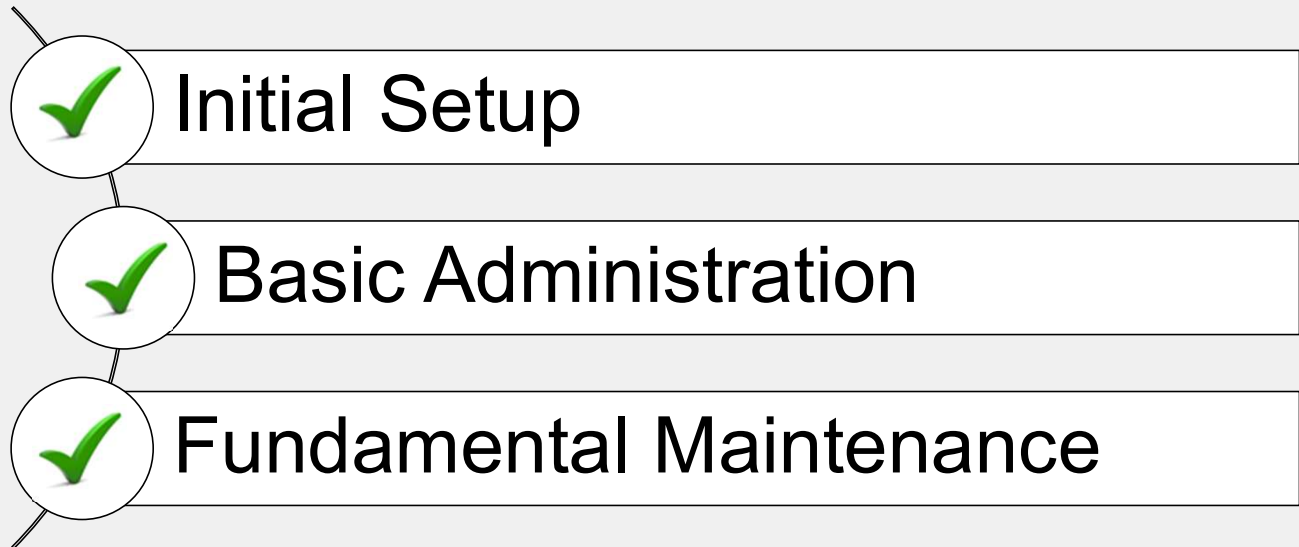
Knowledge Check

1. How do you restrict logins to FortiGate from specific IP addresses?
 - A. Change the FortiGate management interface IP address.
 - ✓ B. Configure a trusted host.

2. When restoring an encrypted system configuration file, in addition to the FortiGate model and firmware version from the time the configuration file was produced, what else must you provide?
 - ✓ A. The password to decrypt the file
 - B. The private decryption key to decrypt the file

3. To increase the chances of success, what document should you consult before upgrading or downgrading firmware?
 - A. *CLI Reference Guide*
 - ✓ B. *FortiOS Release Notes*

Lesson Progress



Congratulations! You have completed this lesson.

Now, you will review the objectives covered in this lesson.

Review

- ✓ Configure FortiGate using the factory default settings
- ✓ Configure FortiGate as the DHCP server
- ✓ Configure and control administrator access to FortiGate
- ✓ Back up and restore system configuration files
- ✓ Upgrade FortiGate firmware
- ✓ Check and verify FortiGuard licenses

This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how and where FortiGate fits into your network and how to perform basic FortiGate administration.